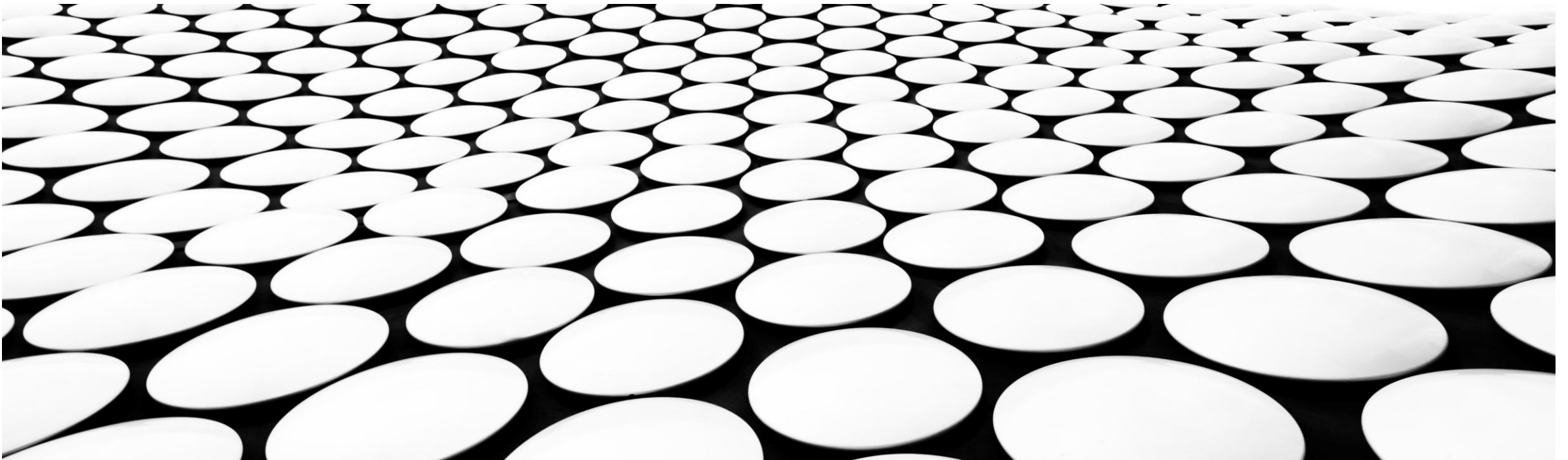


# EXPLORATORY DATA ANALYSIS

*Story behind the data: Titanic dataset*

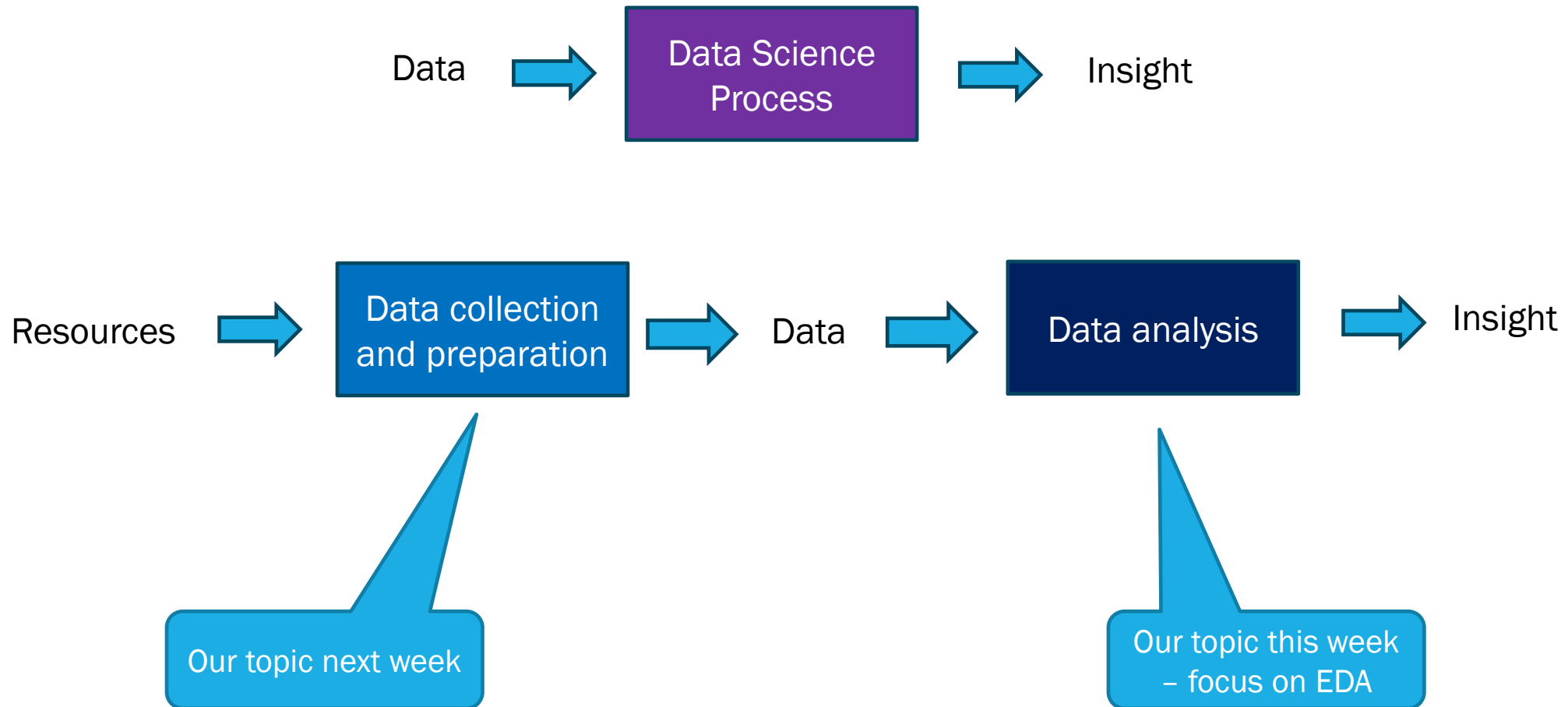




## EXPLORATORY DATA ANALYSIS

- Descriptive statistics
- Diagnostic statistics
- Case Study 1 – Titanic

Friday Case Study 2 +  
Examples for Assignment 1

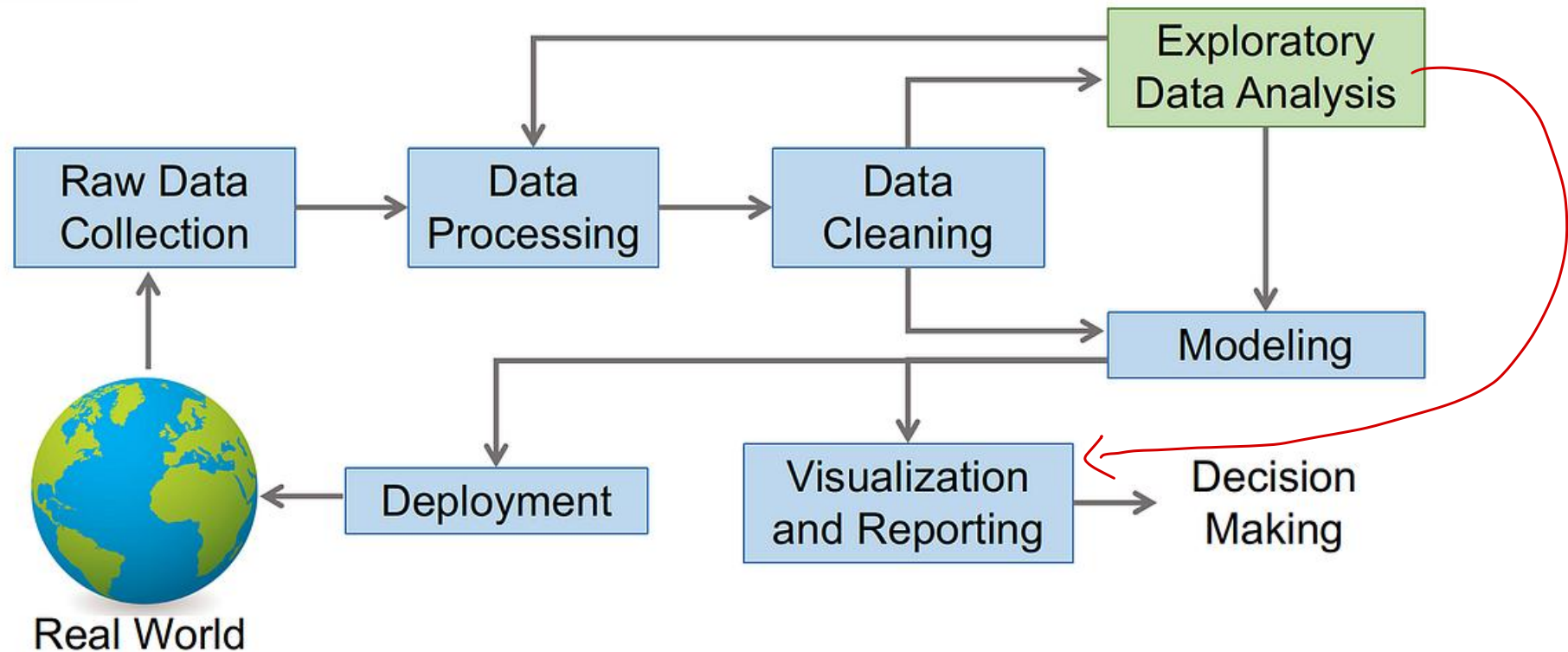


## Data Science Process: A Comprehensive Guide



Abhijit · Follow  
6 min read · Jan 15, 2024

## Data Science Process



## Introduction to Exploratory Data Analysis



Kaushik Mani · Follow

Published in DataDrivenInvestor · 9 min read · Jan 29, 2019

... one of the most important components to any data science experiment that doesn't get as much importance as it should is Exploratory Data Analysis (EDA).

In short, EDA is “A first look at the data”. It is a critical step in analyzing the data from an experiment. It is used to **understand and summarize the content of the dataset** to ensure that the features which we feed to our machine learning algorithms are refined and we get valid, correctly interpreted results.

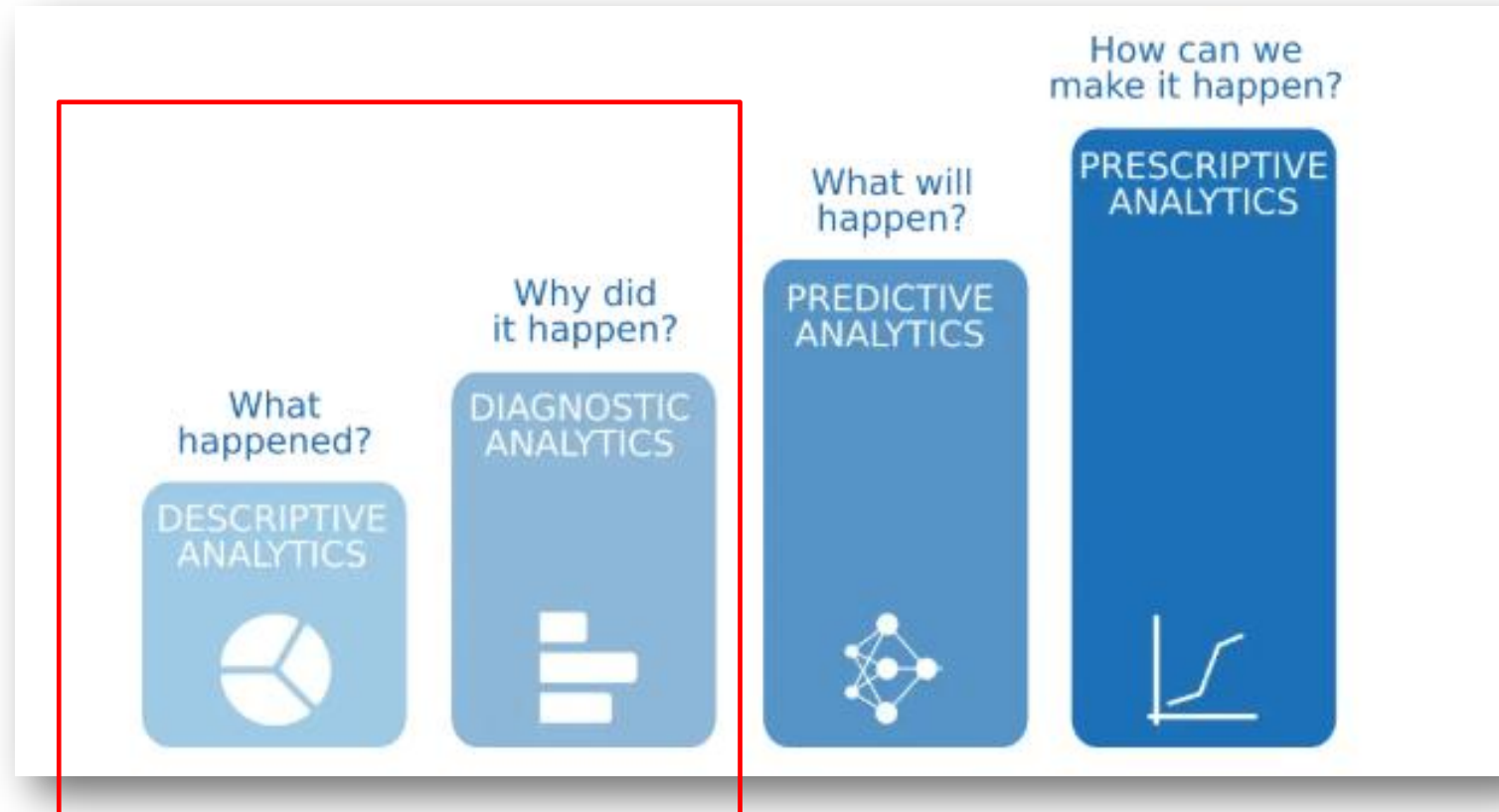
Or that we present valid descriptive insights in reports

# What is advanced analytics?

Tutorials

Pablo Martín

August 31, 2023



# Descriptive Statistics

# Statistics Handbook for Data Analysts



Anita Gupta · Follow  
17 min read · Sep 14, 2024

Statistics Essentials: Understanding  
Key Terms in Data Analysis

## Measures of central tendency

1. Mean
2. Median
3. Mode

## Measures of dispersion

1. Range
2. Variance
3. Standard deviation



**Mean:** The average value.

$$\text{Mean}(\mu) = \frac{1}{n} \sum_{i=1}^n X_i$$

Example

The average salary across United States is \$60,000.

**Median:** The middle value in a sorted dataset. If the dataset contains an even number of values, the median is the average of the two middle values.

### Example

The median divides the data points: 50% are above it, and 50% are below it. So next time someone says the median salary is \$50,000, it means 50% of people earn below that and 50% earn above it.

**Mode:** The most frequently occurring value.

### Example

If you do a poll and ask what food students want to order for lunch, and the majority vote for pizza, then pizza is the mode — the highest occurring value.

# Statistics Handbook for Data Analysts



Anita Gupta · Follow  
17 min read · Sep 14, 2024

Statistics Essentials: Understanding  
Key Terms in Data Analysis

## Measures of central tendency

1. Mean
2. Median
3. Mode

## Measures of dispersion

1. Range
2. Variance
3. Standard deviation

**Range** is from the minimum to the maximum value.

### Example

If you're looking to buy a house, you might be considering houses in the price range of \$500,000 to \$800,000. We use ranges all the time in day-to-day conversations.

**Variance** is a measure of dispersion, meaning it is a measure of how far a set of numbers is spread out from their average value.



$$S^2 = \frac{\sum_{i=1}^n (x_i - \bar{X})^2}{n - 1}$$

$S^2$  = Variance

$n$  = The Number of data Point

$X_i$  = Each of the values of the data

$\bar{X}$  = The Mean of  $X_i$

Equation for a sample

**Standard deviation** is the square root of the variance.

This allows the dispersion metric to be in the same unit as the observed data. For example, we now have dollars instead of dollars squared, making interpretation more useful.



$$S_x = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

$n$  = Number of Observations  
 $x_i$  = Value of the one Observation  
 $\bar{x}$  = Mean Value of Observations

Equation for a  
sample

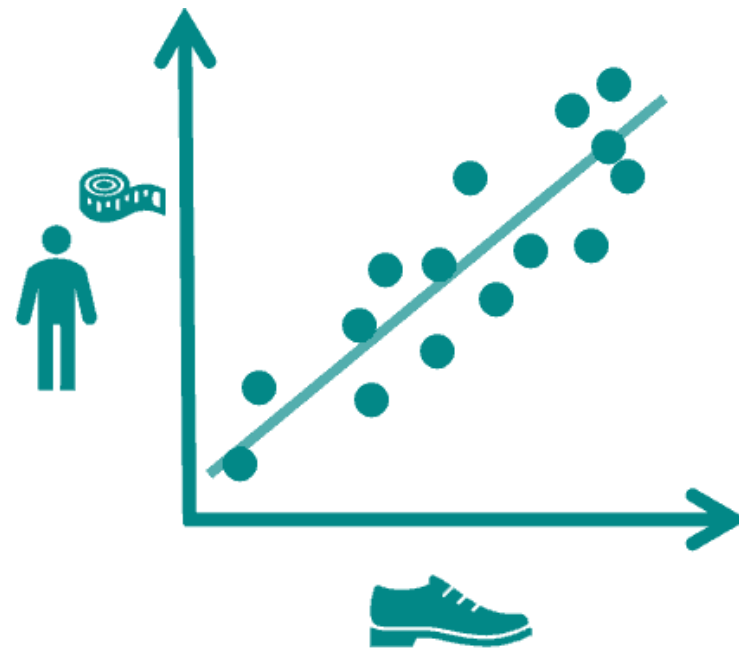
# Diagnostic statistics





# Correlation analysis

Correlation analysis is a statistical method used to evaluate the relationship between two variables, such as the association between body size and shoe size.





## Pearson correlation analysis

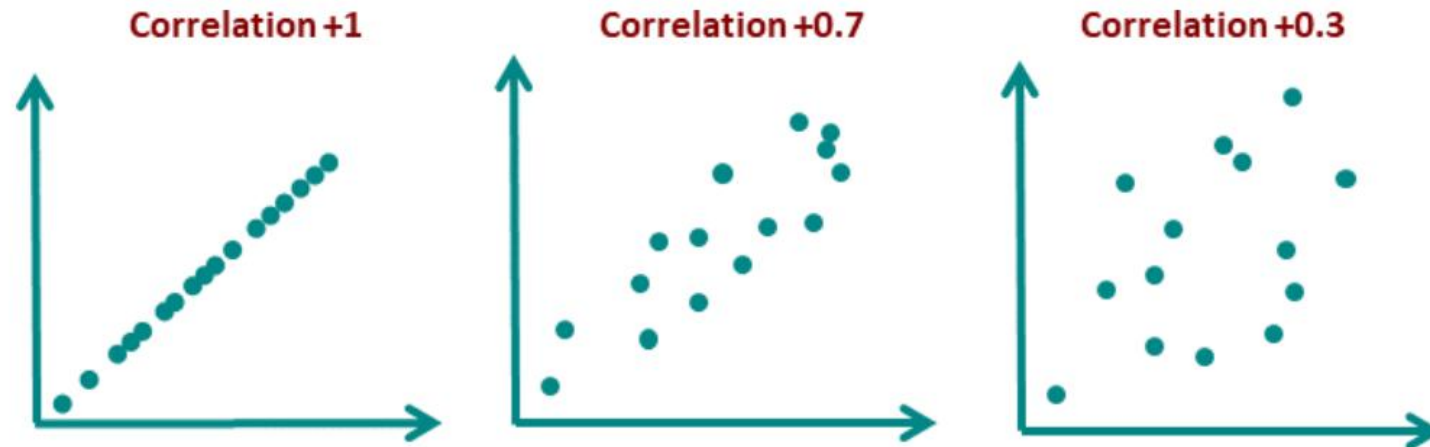
With the **Pearson correlation analysis** you get a statement about the linear correlation between metric scaled variables. The respective **covariance** is used for the calculation.

With the help of correlation analysis two statements can be made:

- one about the direction
- and one about the strength

$$\text{Cov}(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{N - 1}$$

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}}$$

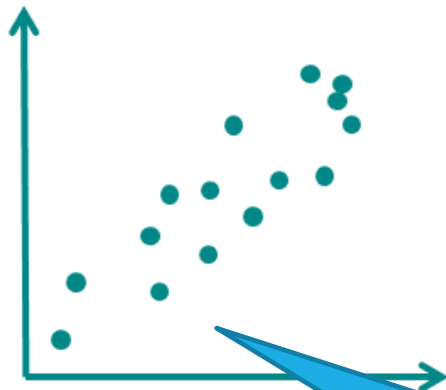


| $ r $       | Strength of correlation |
|-------------|-------------------------|
| $0.0 < 0.1$ | no correlation          |
| $0.1 < 0.3$ | little correlation      |
| $0.3 < 0.5$ | medium correlation      |
| $0.5 < 0.7$ | high correlation        |
| $0.7 < 1$   | very high correlation   |

# Requirements for causality

1

There is a significant correlation



2

Timely

Variable A surveyed



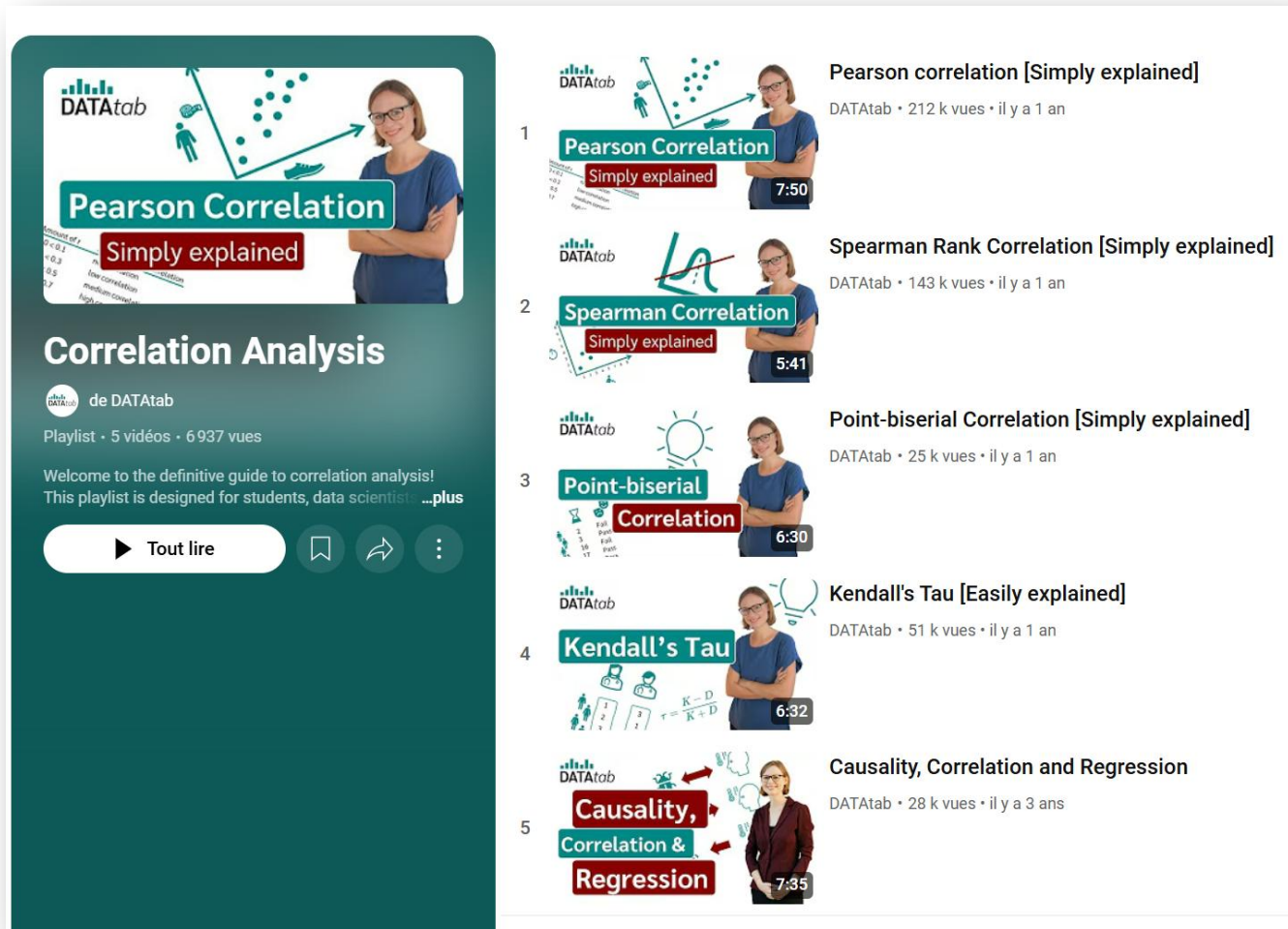
Variable B surveyed

Theory



Profound theory

If we don't even have correlation...  
for sure we can affirm there is NO  
causality



**Pearson Correlation**  
Simply explained

## Correlation Analysis

de DATAtab

Playlist • 5 vidéos • 6 937 vues

Welcome to the definitive guide to correlation analysis!  
This playlist is designed for students, data scientists, ...plus

Tout lire

- Pearson Correlation**  
Simply explained  
7:50  
DATAtab • 212 k vues • il y a 1 an
- Spearman Correlation**  
Simply explained  
5:41  
DATAtab • 143 k vues • il y a 1 an
- Point-biserial Correlation**  
Simply explained  
6:30  
DATAtab • 25 k vues • il y a 1 an
- Kendall's Tau**  
Easily explained  
6:32  
DATAtab • 51 k vues • il y a 1 an
- Causality, Correlation and Regression**  
7:35  
DATAtab • 28 k vues • il y a 3 ans

Choice varies based on conditions (parametric: metric data normally distributed)

# *Case Study 1*

## Titanic Dataset

```
# Import libraries for visualization
import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Loading the dataset
titanic = sns.load_dataset('titanic')
```

```
# Number of samples, Number of attributes
titanic.shape
```

```
(891, 15)
```

Dataset available  
within seaborn

Lots of other titanic  
datasets!!



2,288 Datasets



#### Titanic

[Khashayar Baghizadeh](#) · Updated 8 years ago  
Usability 7.1 · 1 File (CSV) · 11 kB



#### Titanic dataset

[Brenda N](#) · Updated 3 years ago  
Usability 10.0 · 1 File (CSV) · 12 kB



#### Titanic

[Azeem Bootwala](#) · Updated 8 years ago  
Usability 8.2 · 2 Files (CSV) · 12 kB



#### Titanic Dataset

[M Yasser H](#) · Updated 3 years ago  
Usability 10.0 · 1 File (CSV) · 23 kB



#### Titanic

[Rahul](#) · Updated 5 years ago  
Usability 6.8 · 3 Files (CSV) · 35 kB



#### Titanic Dataset

[Shubham\\_Gupta012](#) · Updated 2 years ago  
Usability 10.0 · 1 File (CSV) · 7 kB

```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column      Non-Null Count  Dtype
---  -
0   survived    891 non-null    int64
1   pclass      891 non-null    int64
2   sex         891 non-null    object
3   age         714 non-null    float64
4   sibsp       891 non-null    int64
5   parch       891 non-null    int64
6   fare        891 non-null    float64
7   embarked    889 non-null    object
8   class       891 non-null    category
9   who         891 non-null    object
10  adult_male   891 non-null    bool
11  deck        203 non-null    category
12  embark_town  889 non-null    object
13  alive       891 non-null    object
14  alone       891 non-null    bool
```

A first view of attributes and data types

#### Siblings (sibsp)

- Brother, sister
- Step-brother, step-sister
- Husband, wife

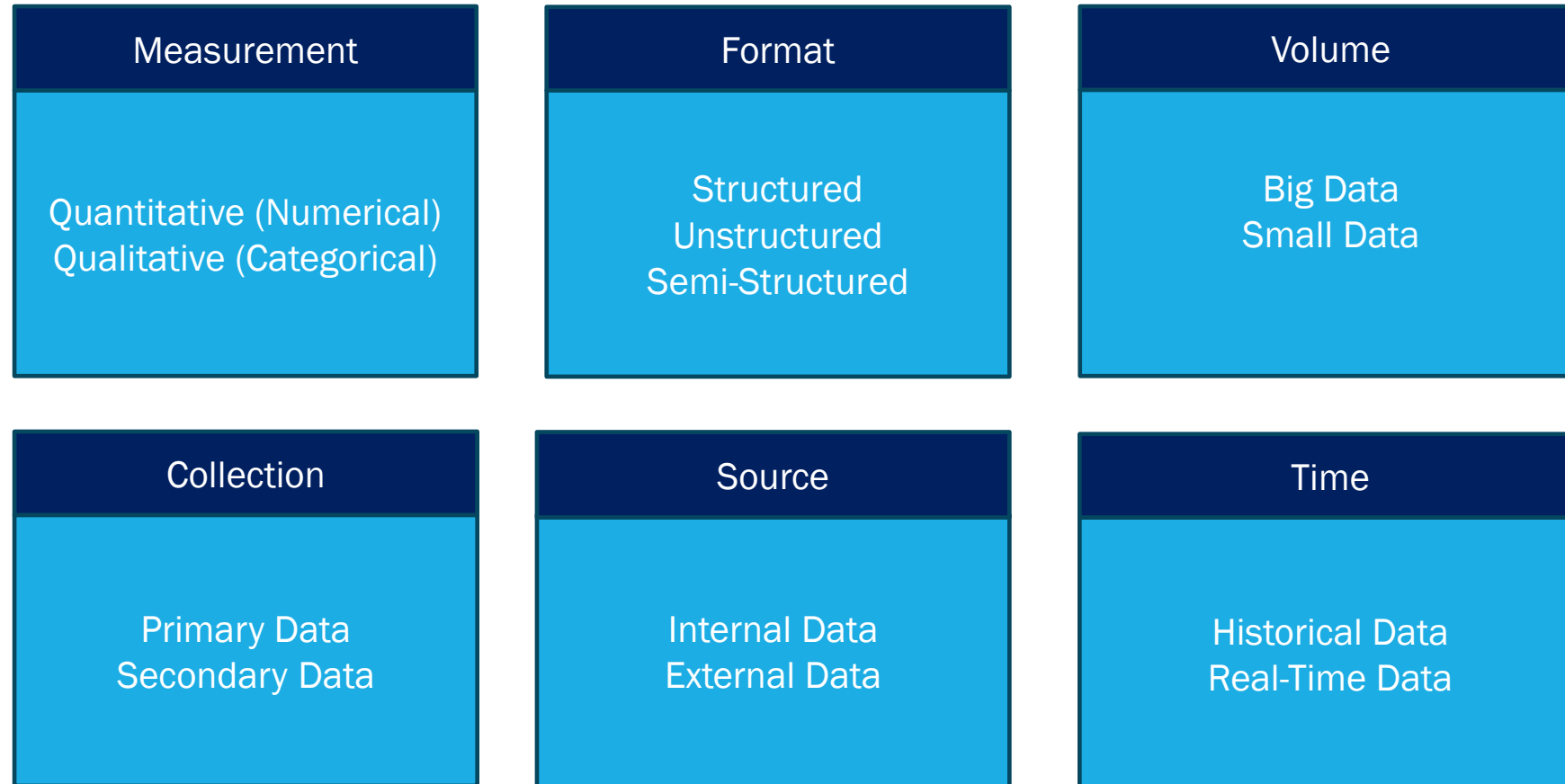
#### Parents/Children (parch)

- Mother, father
- Son, daughter
- Step-children



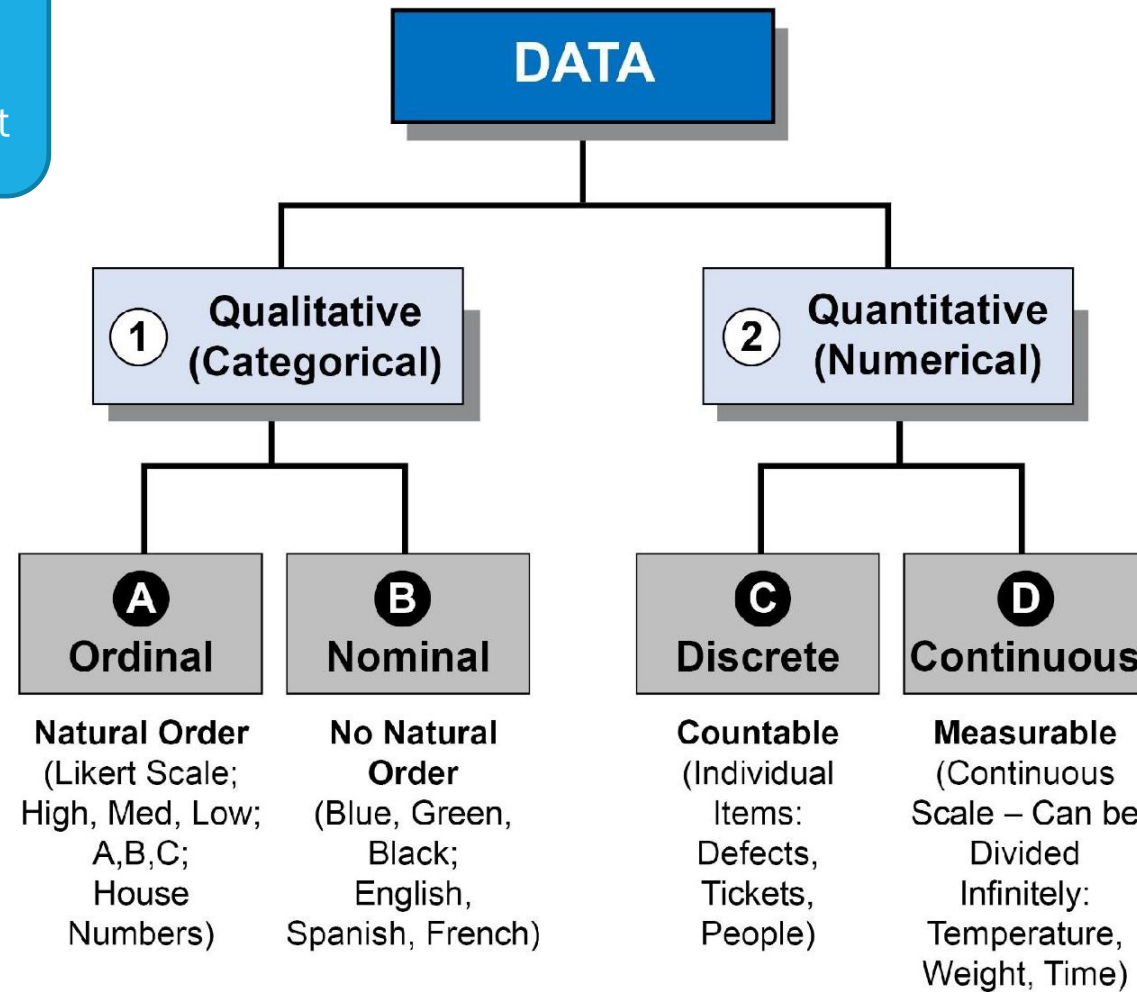
▶ titanic.head(10)

| ... | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class  | who   | adult_male | deck | embark_town | alive | alone |
|-----|----------|--------|--------|------|-------|-------|---------|----------|--------|-------|------------|------|-------------|-------|-------|
| 0   | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third  | man   | True       | NaN  | Southampton | no    | False |
| 1   | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First  | woman | False      | C    | Cherbourg   | yes   | False |
| 2   | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third  | woman | False      | NaN  | Southampton | yes   | True  |
| 3   | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First  | woman | False      | C    | Southampton | yes   | False |
| 4   | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | NaN  | Southampton | no    | True  |
| 5   | 0        | 3      | male   | NaN  | 0     | 0     | 8.4583  | Q        | Third  | man   | True       | NaN  | Queenstown  | no    | True  |
| 6   | 0        | 1      | male   | 54.0 | 0     | 0     | 51.8625 | S        | First  | man   | True       | E    | Southampton | no    | True  |
| 7   | 0        | 3      | male   | 2.0  | 3     | 1     | 21.0750 | S        | Third  | child | False      | NaN  | Southampton | no    | False |
| 8   | 1        | 3      | female | 27.0 | 0     | 2     | 11.1333 | S        | Third  | woman | False      | NaN  | Southampton | yes   | False |
| 9   | 1        | 2      | female | 14.0 | 1     | 0     | 30.0708 | C        | Second | child | False      | NaN  | Cherbourg   | yes   | False |



How would we  
characterize our  
current dataset?

We saw last week that this dimension for data characterization was important



▶ titanic.head(10)

| ... | survived | pclass | sex    | age  | sibsp | parch | fare    | embarked | class  | who   | adult_male | deck | embark_town | alive | alone |
|-----|----------|--------|--------|------|-------|-------|---------|----------|--------|-------|------------|------|-------------|-------|-------|
| 0   | 0        | 3      | male   | 22.0 | 1     | 0     | 7.2500  | S        | Third  | man   | True       | NaN  | Southampton | no    | False |
| 1   | 1        | 1      | female | 38.0 | 1     | 0     | 71.2833 | C        | First  | woman | False      | C    | Cherbourg   | yes   | False |
| 2   | 1        | 3      | female | 26.0 | 0     | 0     | 7.9250  | S        | Third  | woman | False      | NaN  | Southampton | yes   | True  |
| 3   | 1        | 1      | female | 35.0 | 1     | 0     | 53.1000 | S        | First  | woman | False      | C    | Southampton | yes   | False |
| 4   | 0        | 3      | male   | 35.0 | 0     | 0     | 8.0500  | S        | Third  | man   | True       | NaN  | Southampton | no    | True  |
| 5   | 0        | 3      | male   | NaN  | 0     | 0     | 8.4583  | Q        | Third  | man   | True       | NaN  | Queenstown  | no    | True  |
| 6   | 0        | 1      | male   | 54.0 | 0     | 0     | 51.8625 | S        | First  | man   | True       | E    | Southampton | no    | True  |
| 7   | 0        | 3      | male   | 2.0  | 3     | 1     | 21.0750 | S        | Third  | child | False      | NaN  | Southampton | no    | False |
| 8   | 1        | 3      | female | 27.0 | 0     | 2     | 11.1333 | S        | Third  | woman | False      | NaN  | Southampton | yes   | False |
| 9   | 1        | 2      | female | 14.0 | 1     | 0     | 30.0708 | C        | Second | child | False      | NaN  | Cherbourg   | yes   | False |

## Feature analysis

```
titanic.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 891 entries, 0 to 890  
Data columns (total 15 columns):  
#   Column      Non-Null Count  Dtype  
---  -  
0   survived    891 non-null    int64  
1   pclass      891 non-null    int64  
2   sex         891 non-null    object  
3   age         714 non-null    float64  
4   sibsp       891 non-null    int64  
5   parch       891 non-null    int64  
6   fare        891 non-null    float64  
7   embarked    889 non-null    object  
8   class       891 non-null    category  
9   who         891 non-null    object  
10  adult_male   891 non-null    bool  
11  deck        203 non-null    category  
12  embark_town  889 non-null    object  
13  alive        891 non-null    object  
14  alone        891 non-null    bool
```

A view at missing data also...  
we'll do data cleaning later in  
the semester

A view at perhaps redundant  
attributes?

Importance of self-  
explanatory labels when  
documentation is lost...

## Descriptive statistics

```
titanic.describe()
```

|              | survived   | pclass     | age        | sibsp      | parch      | fare       |
|--------------|------------|------------|------------|------------|------------|------------|
| <b>count</b> | 891.000000 | 891.000000 | 714.000000 | 891.000000 | 891.000000 | 891.000000 |
| <b>mean</b>  | 0.383838   | 2.308642   | 29.699118  | 0.523008   | 0.381594   | 32.204208  |
| <b>std</b>   | 0.486592   | 0.836071   | 14.526497  | 1.102743   | 0.806057   | 49.693429  |
| <b>min</b>   | 0.000000   | 1.000000   | 0.420000   | 0.000000   | 0.000000   | 0.000000   |
| <b>25%</b>   | 0.000000   | 2.000000   | 20.125000  | 0.000000   | 0.000000   | 7.910400   |
| <b>50%</b>   | 0.000000   | 3.000000   | 28.000000  | 0.000000   | 0.000000   | 14.454200  |
| <b>75%</b>   | 1.000000   | 3.000000   | 38.000000  | 1.000000   | 0.000000   | 31.000000  |
| <b>max</b>   | 1.000000   | 3.000000   | 80.000000  | 8.000000   | 6.000000   | 512.329200 |

Central tendency

Dispersion

```
▶ titanic.describe(include=["object", "category"])
```

...

|               | sex  | embarked | class | who | deck | embark_town | alive |
|---------------|------|----------|-------|-----|------|-------------|-------|
| <b>count</b>  | 891  | 889      | 891   | 891 | 203  | 889         | 891   |
| <b>unique</b> | 2    | 3        | 3     | 3   | 7    | 3           | 2     |
| <b>top</b>    | male | S        | Third | man | C    | Southampton | no    |
| <b>freq</b>   | 577  | 644      | 491   | 537 | 59   | 644         | 549   |

Mode is on  
categorical data

Dispersion on  
categorical data

```
▶ titanic.value_counts('embark_town')
```

| embark_town | count |
|-------------|-------|
| Southampton | 644   |
| Cherbourg   | 168   |
| Queenstown  | 77    |

```
▶ titanic.value_counts('class')
```

| class  | count |
|--------|-------|
| Third  | 491   |
| First  | 216   |
| Second | 184   |

```
▶ titanic.value_counts('who')
```

| who   | count |
|-------|-------|
| man   | 537   |
| woman | 271   |
| child | 83    |



```
▶ titanic.value_counts('class')
```

| ...    |  | count |
|--------|--|-------|
| class  |  |       |
| Third  |  | 491   |
| First  |  | 216   |
| Second |  | 184   |

```
▶ titanic.value_counts('pclass')
```

| ...    |  | count |
|--------|--|-------|
| pclass |  |       |
| 3      |  | 491   |
| 1      |  | 216   |
| 2      |  | 184   |

Redundancy

Sometimes a numerical feature is really a categorical feature

```
titanic[["age"]].describe()
```

| age   |            |
|-------|------------|
| count | 714.000000 |
| mean  | 29.699118  |
| std   | 14.526497  |
| min   | 0.420000   |
| 25%   | 20.125000  |
| 50%   | 28.000000  |
| 75%   | 38.000000  |
| max   | 80.000000  |

```
titanic.value_counts('age')
```

| age   |     |
|-------|-----|
| count |     |
| 24.0  | 30  |
| 22.0  | 27  |
| 18.0  | 26  |
| 19.0  | 25  |
| 30.0  | 25  |
| ...   | ... |
| 53.0  | 1   |
| 66.0  | 1   |
| 70.5  | 1   |
| 74.0  | 1   |
| 80.0  | 1   |

In general value\_counts on a numerical data does not make sense

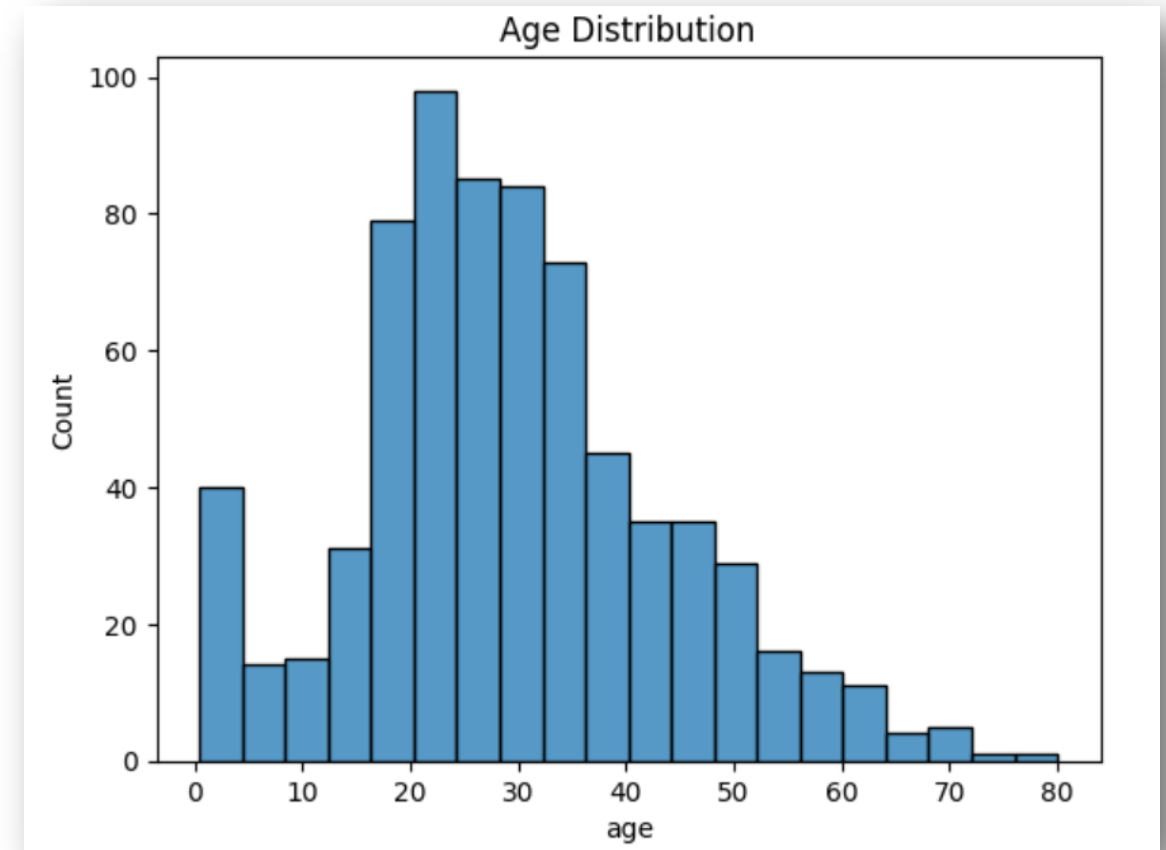
Better with graphs for visualisation of dispersion

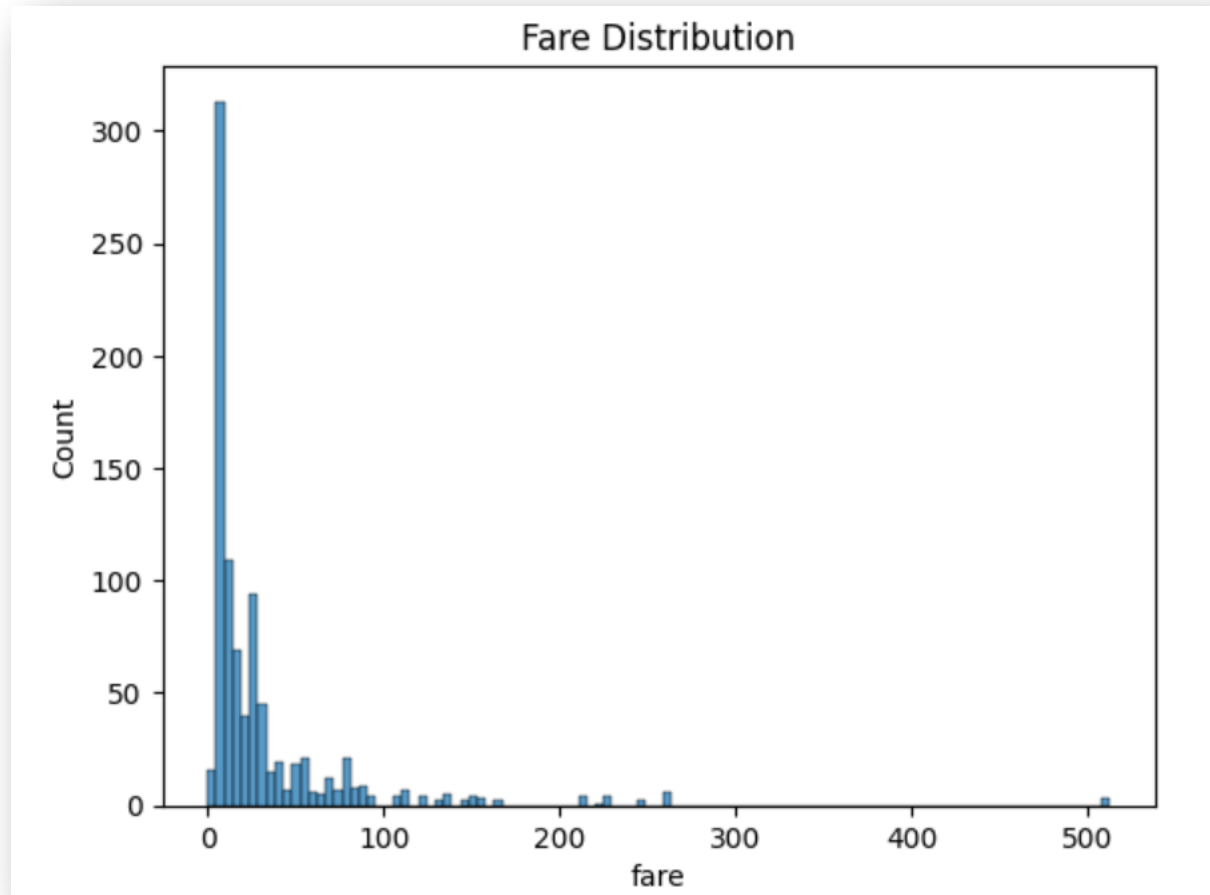
## Univariate analysis Numerical Data

### Histograms

A histogram is a bar chart that displays the frequency of a numerical variable's values. It is created by dividing the data into intervals, called bins, and counting the number of observations that fall within each bin.

```
▶ sns.histplot(data=titanic, x="age")  
  plt.title("Age Distribution")  
  plt.show()
```





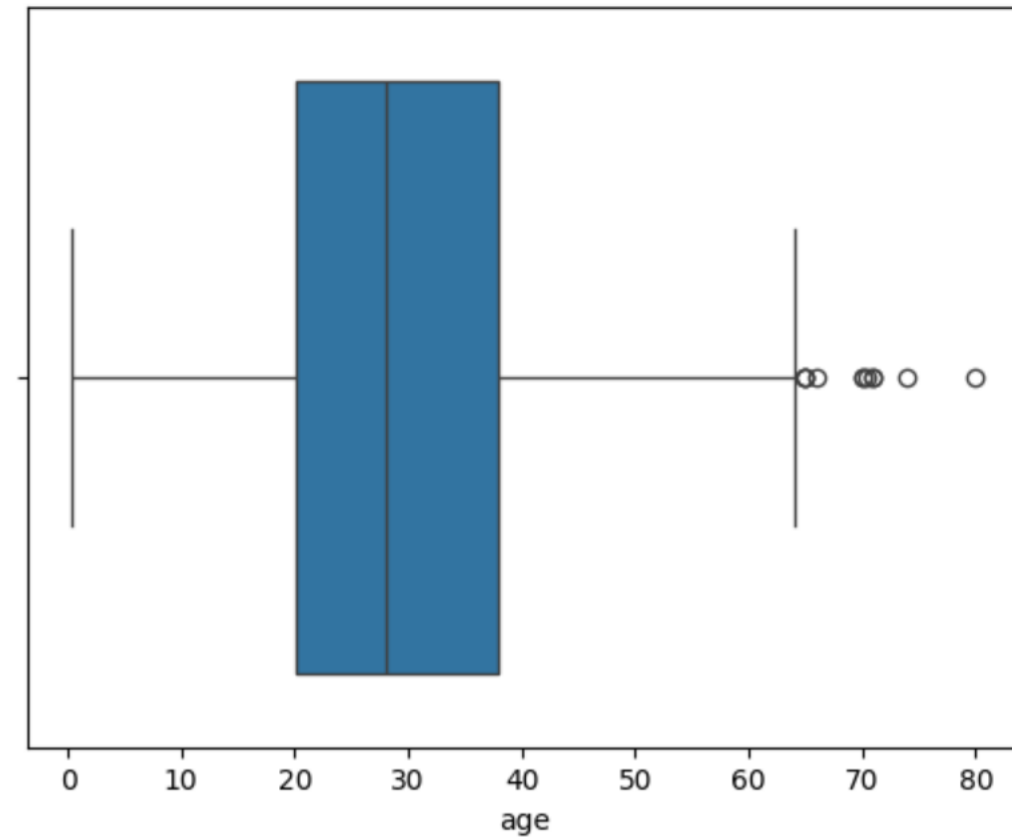
Further study of  
distributions...

Perhaps an outlier?

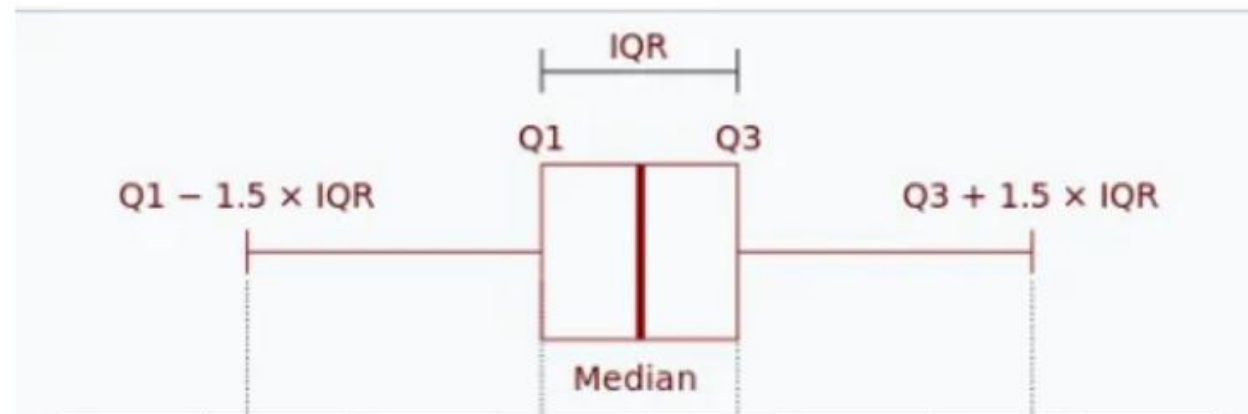
## Box Plots

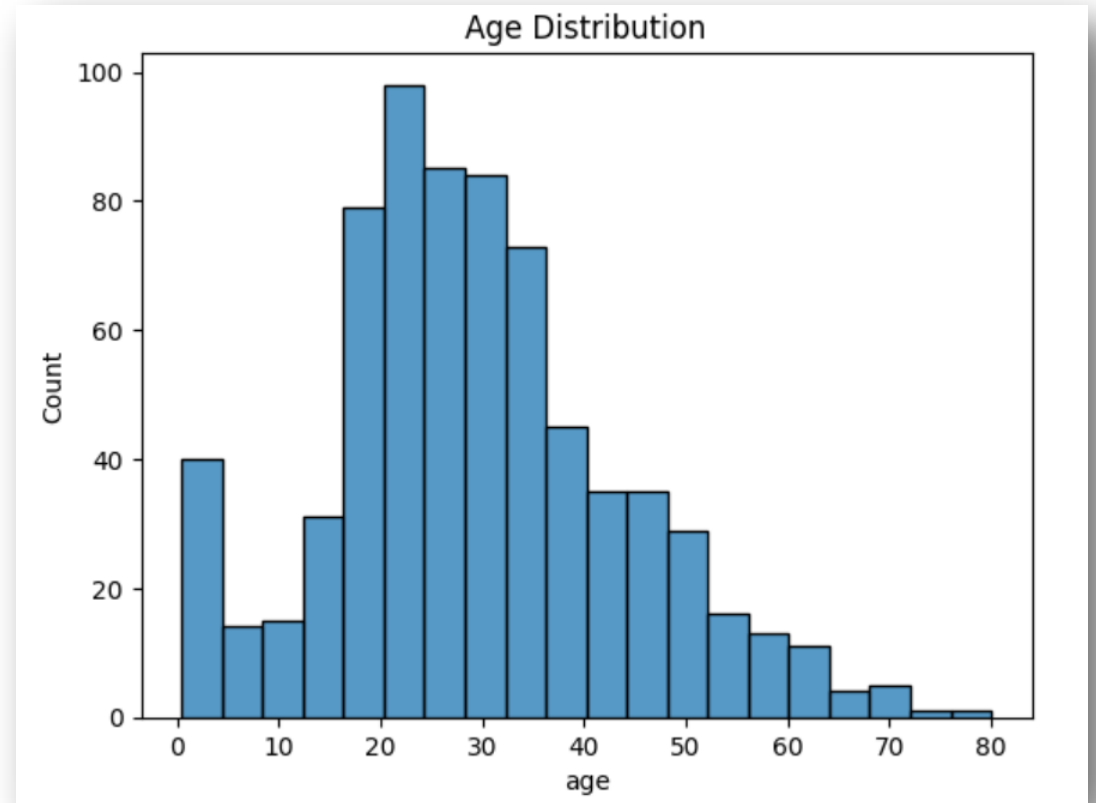
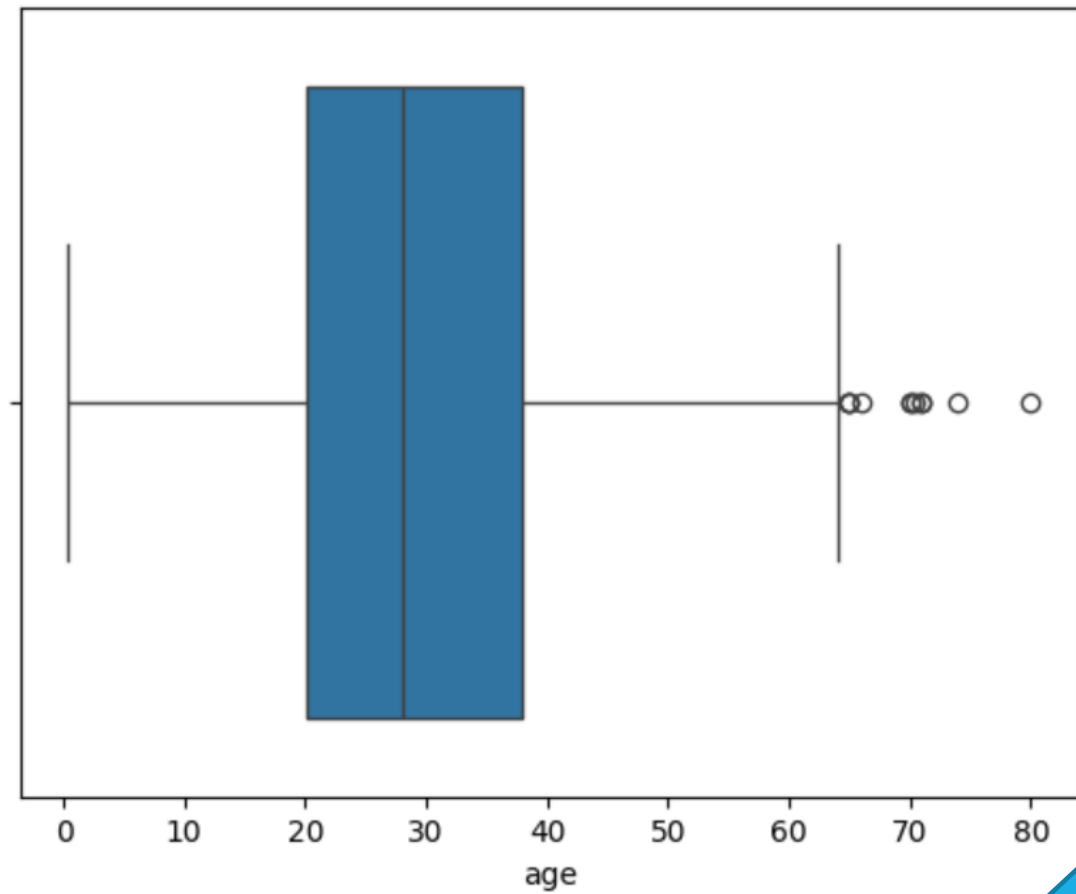
A box plot is a graphical summary of a numerical distribution showing its median, quartiles, spread, and potential outliers.

```
sns.boxplot(x="age", data=titanic)
```



- The **middle line** represents the **median**, which means that 50% of the data points lie below this line, and 50% lie above.
- Then we have the **first quartile (Q1)** and the **third quartile (Q3)**. The median is the second quartile.
- The range between Q1 and Q3 is called the **interquartile range (IQR)**, which represents 50% of the data.
- The **whiskers** extend from Q1 and Q3 to the **minimum** and **maximum** values within a defined range.





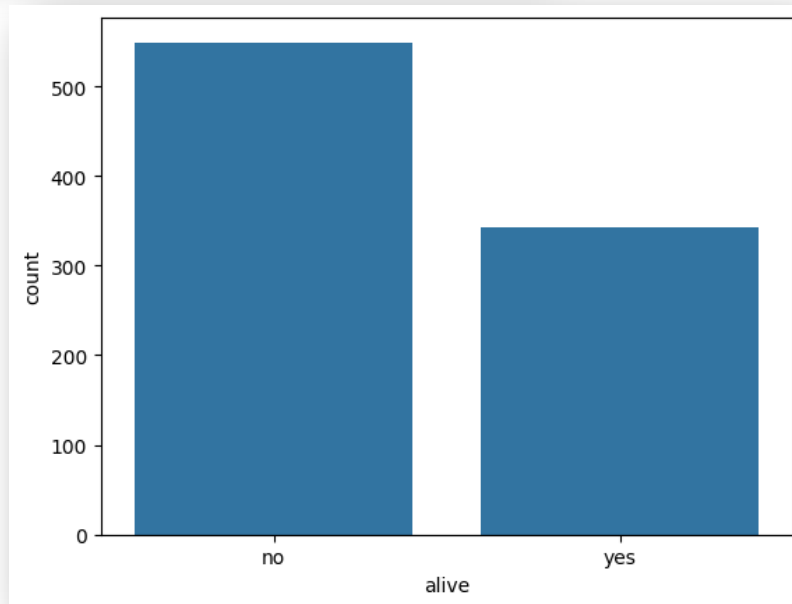
Often used side by side

## Univariate analysis Categorical Data

### Count Plots

A count plot is a type of bar plot that displays the count of observations in each category.

```
sns.countplot(x="alive", data=titanic)  
plt.show()
```



Data obtained about 891 passengers on the Titanic shows that most passengers died in the incident.

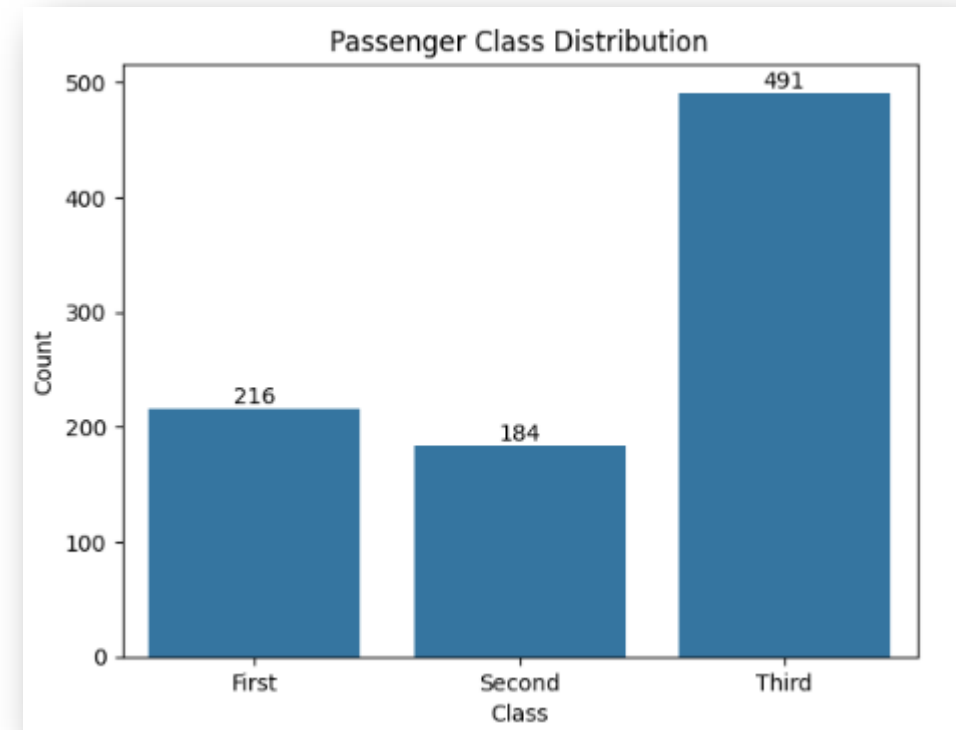
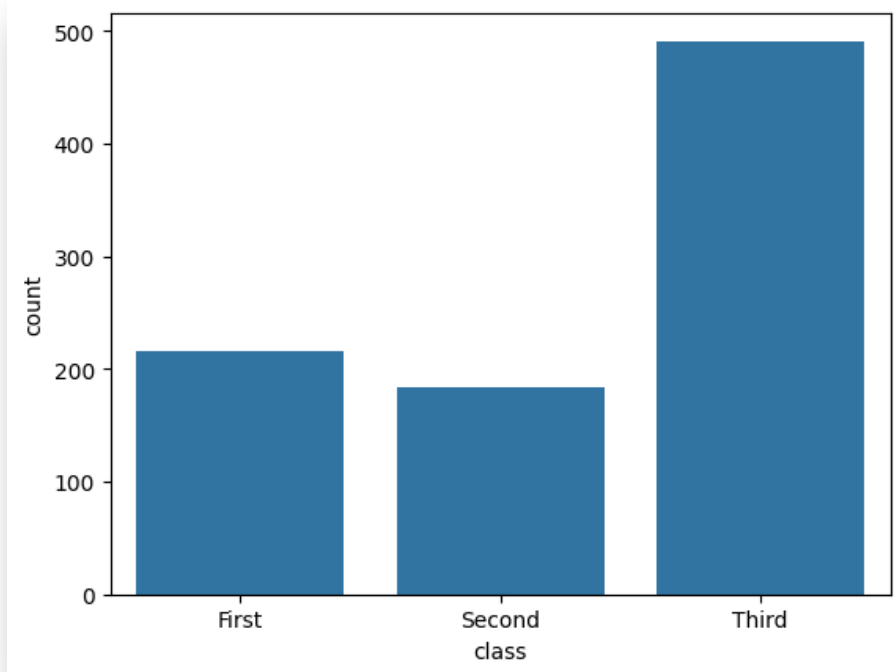




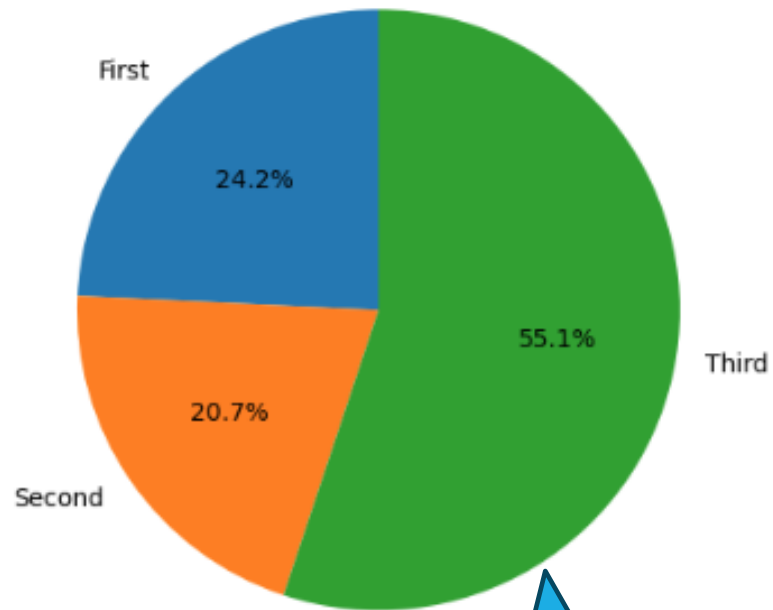
## Communication Skills



Good practice to write a sentence (insight) from what you observe in the graph.

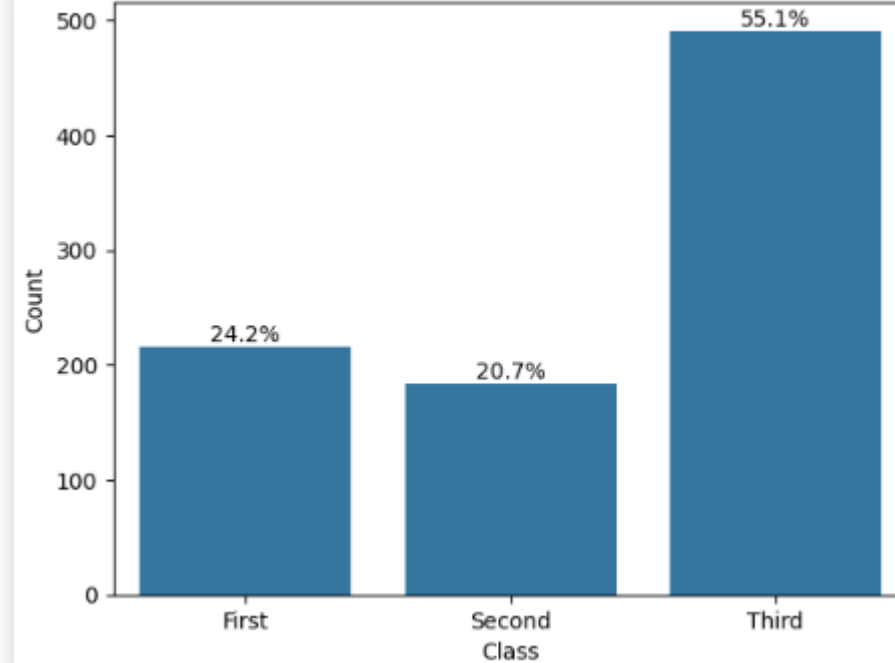


Passenger Class Distribution on the Titanic

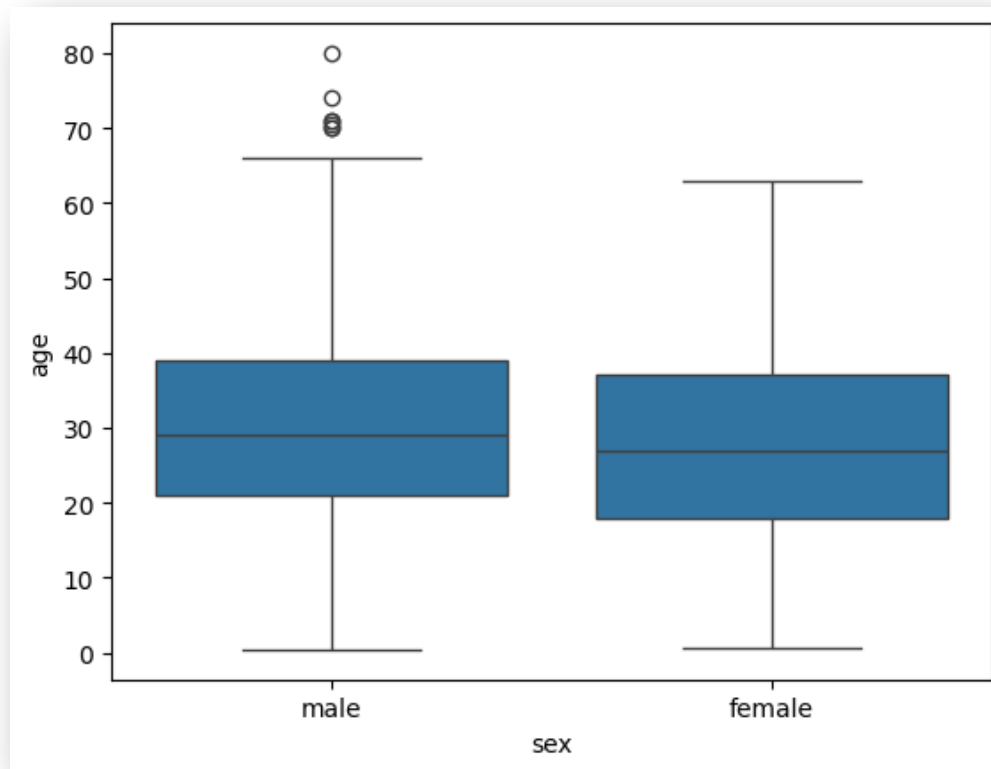


What about Pie Chart?

Passenger Class Distribution (%)



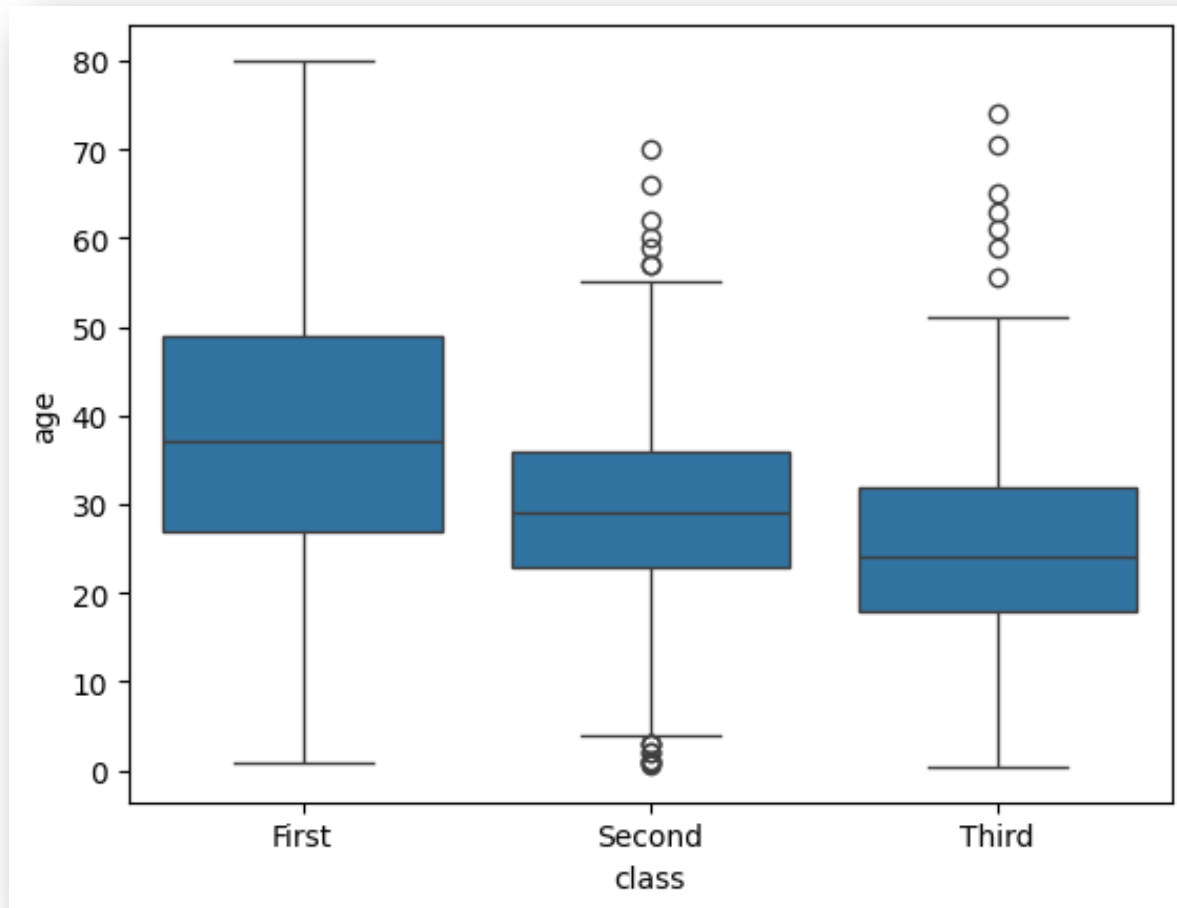
## Bivariate analysis Numerical/Categorical



Shows central tendency statistics of one variable according to values of a second variable

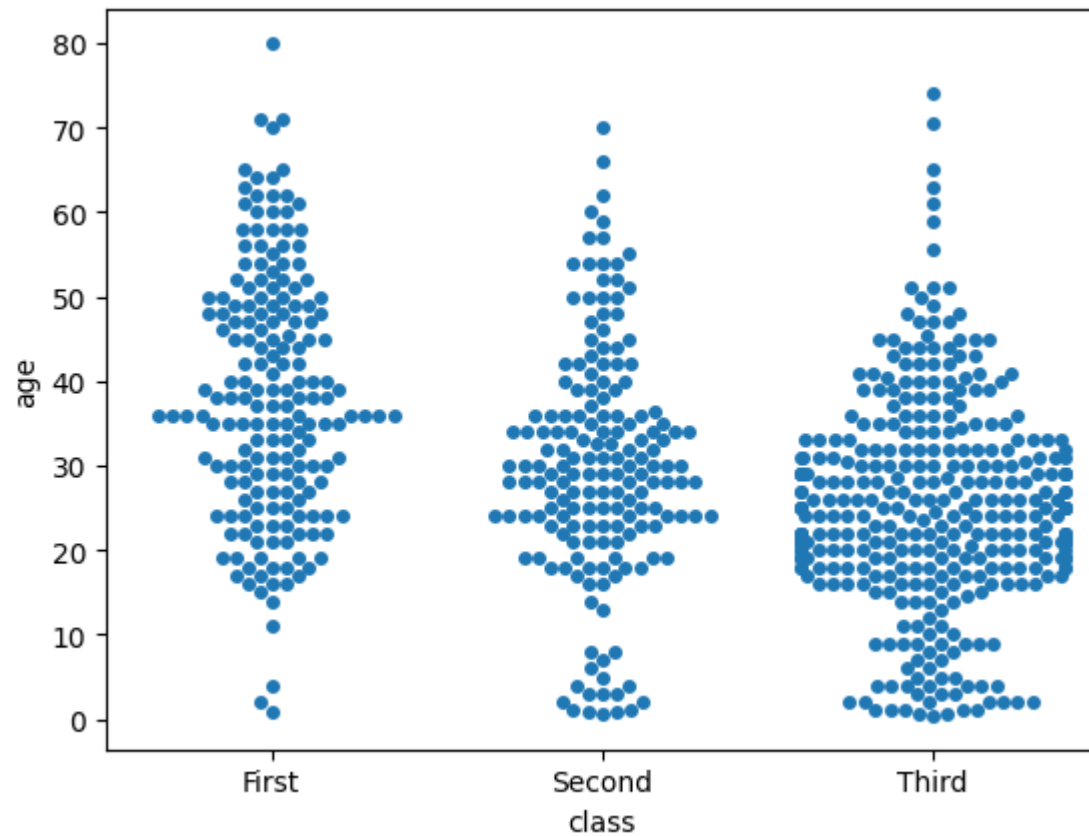
```
sns.boxplot(x="sex", y="age", data=titanic)
```

```
sns.boxplot(x="class", y="age", data=titanic)
```

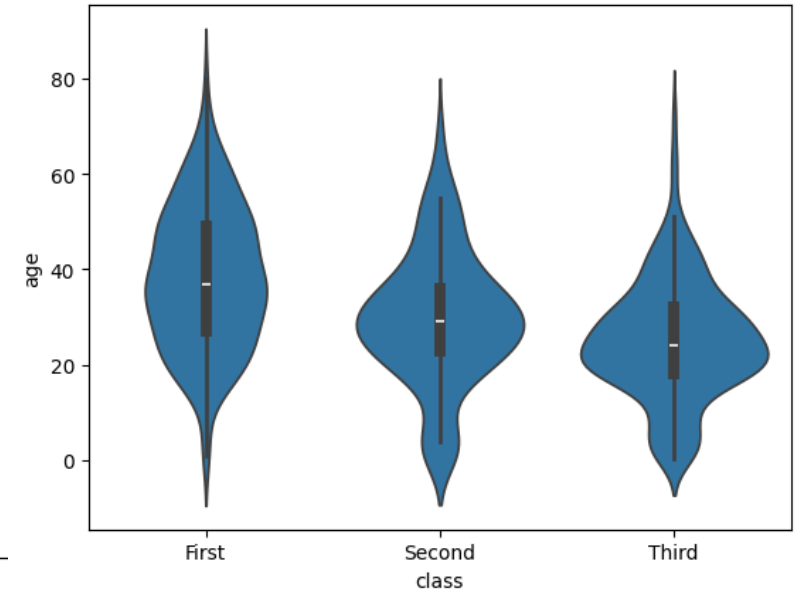
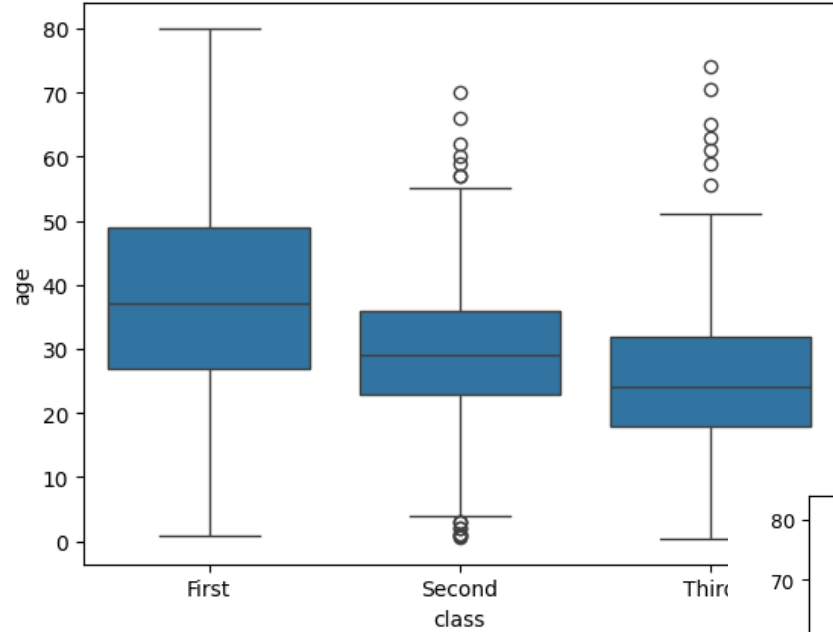


Insight?

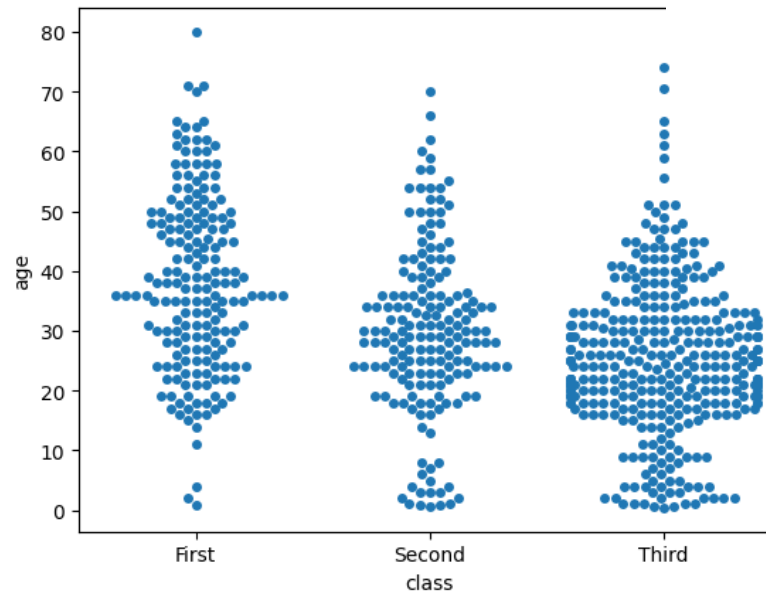
```
sns.swarmplot(x="class", y="age", data=titanic)
```



What do you think?



```
sns.violinplot(x="class", y="age", data=titanic)
```

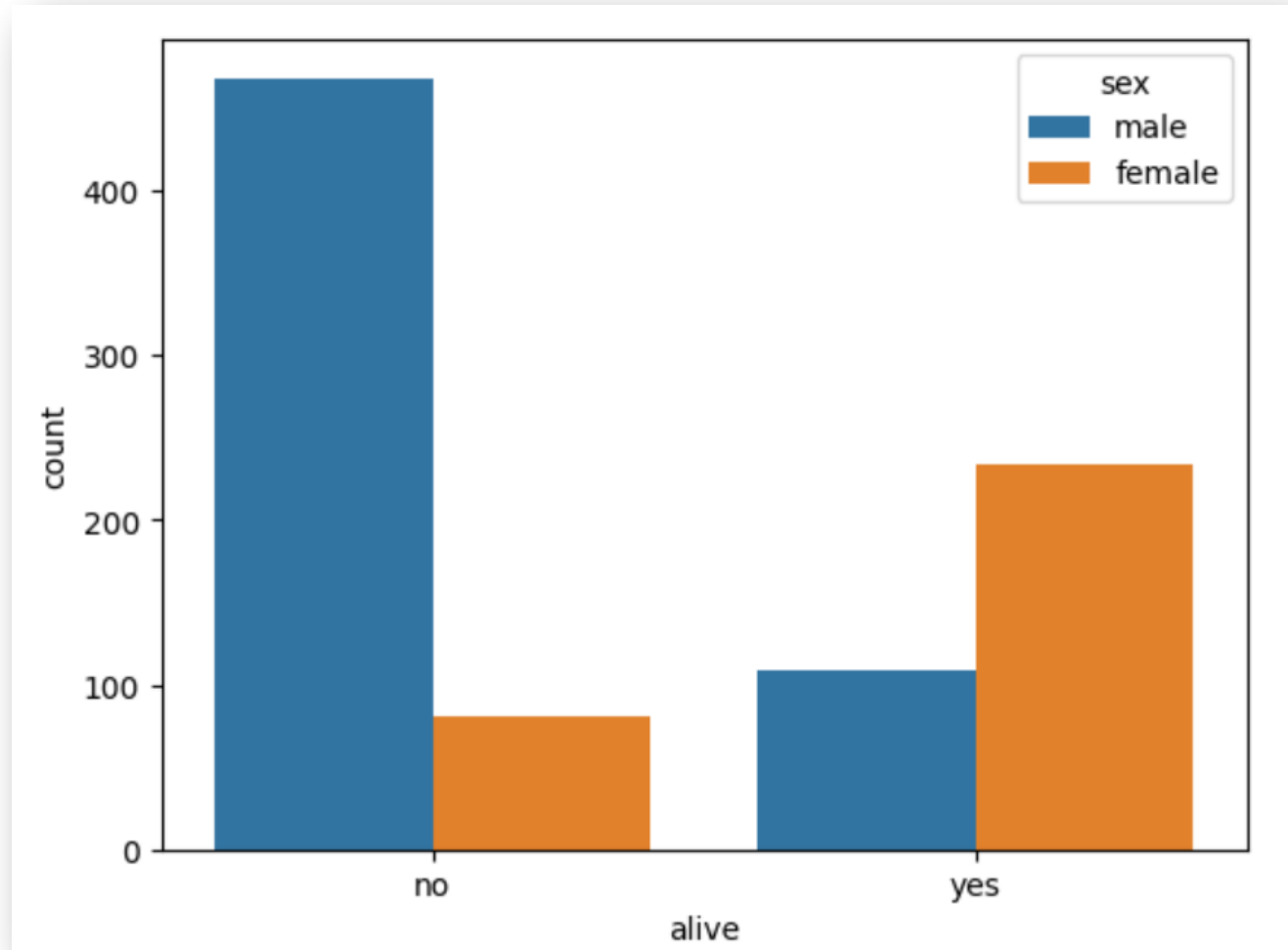


```
sns.swarmplot(x="class", y="age", data=titanic)
```

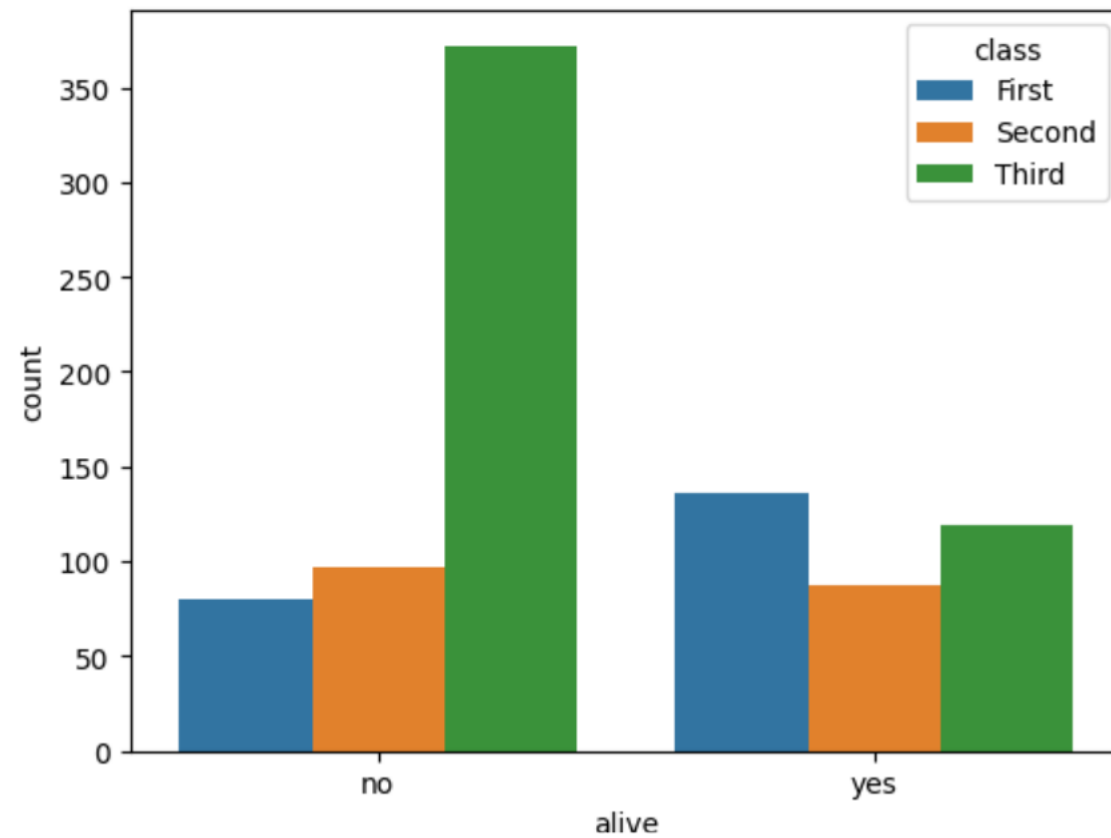
Which would you put in the report? Only a visual (preference) difference?

Bivariate analysis  
Categorical/Categorical

```
sns.countplot(x="alive", hue="sex", data=titanic)  
plt.show()
```

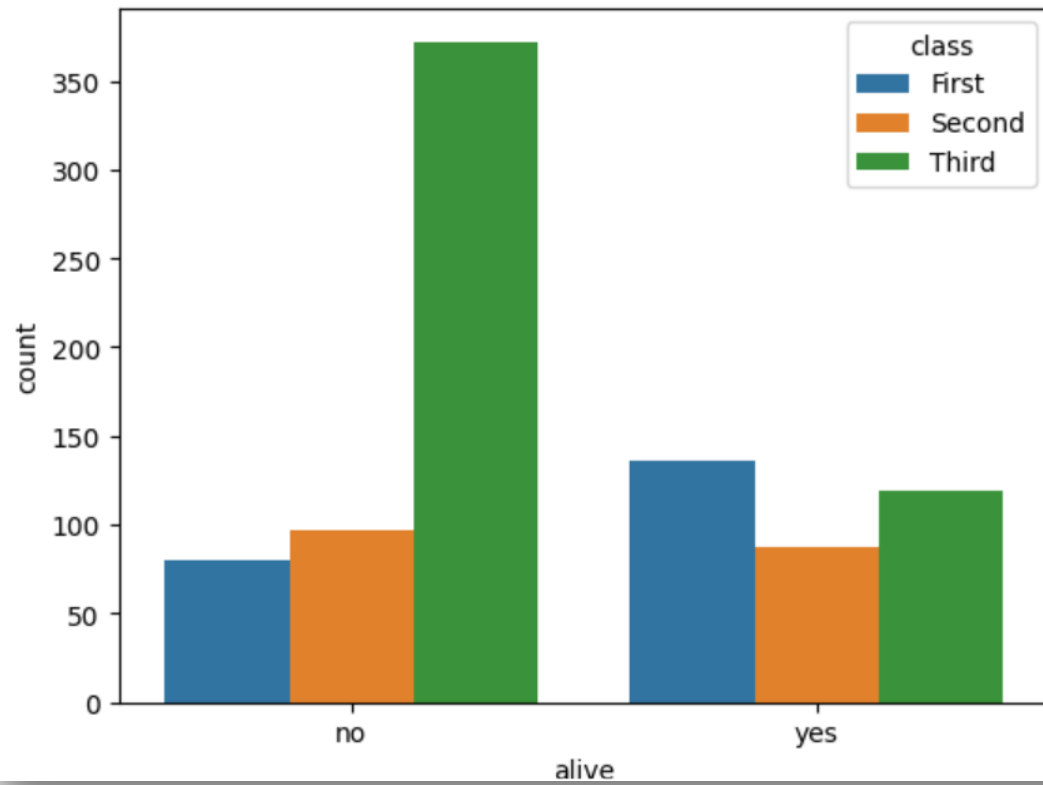


```
sns.countplot(x="alive", hue="class", data=titanic)  
plt.show()
```

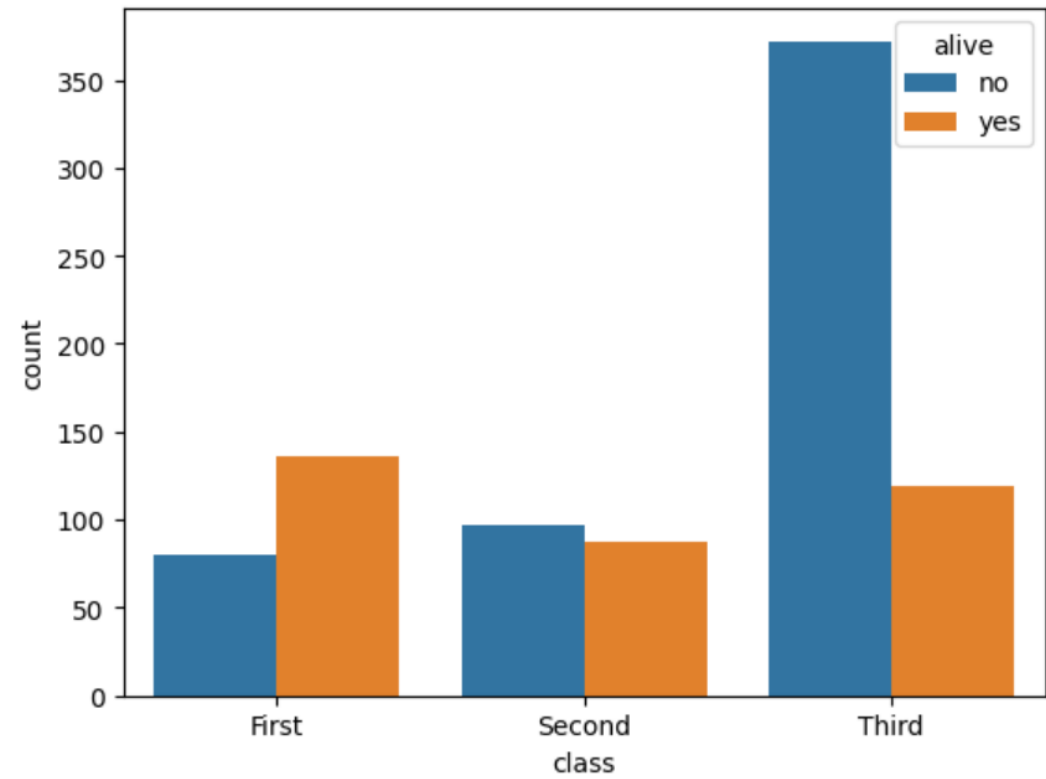




```
sns.countplot(x="alive", hue="class", data=titanic)  
plt.show()
```

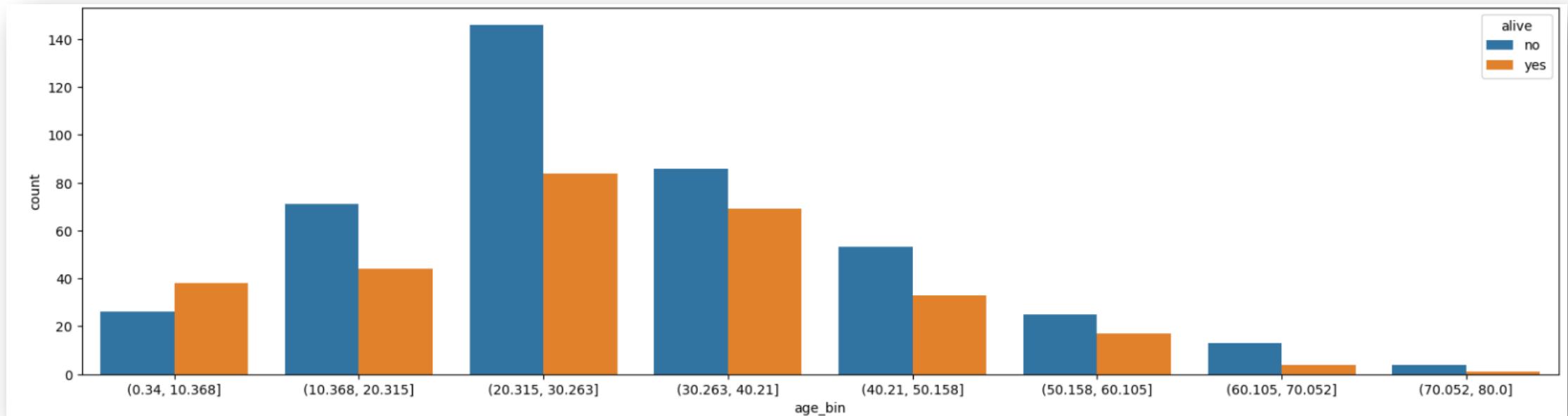


```
sns.countplot(x="class", hue="alive", data=titanic)  
plt.show()
```



## Bivariate analysis Categorical/Numerical

```
titanic['age_bin'] = pd.cut(titanic['age'], 8)  
fig = plt.figure(figsize=(20,5))  
sns.countplot(x="age_bin", hue="alive", data=titanic)
```



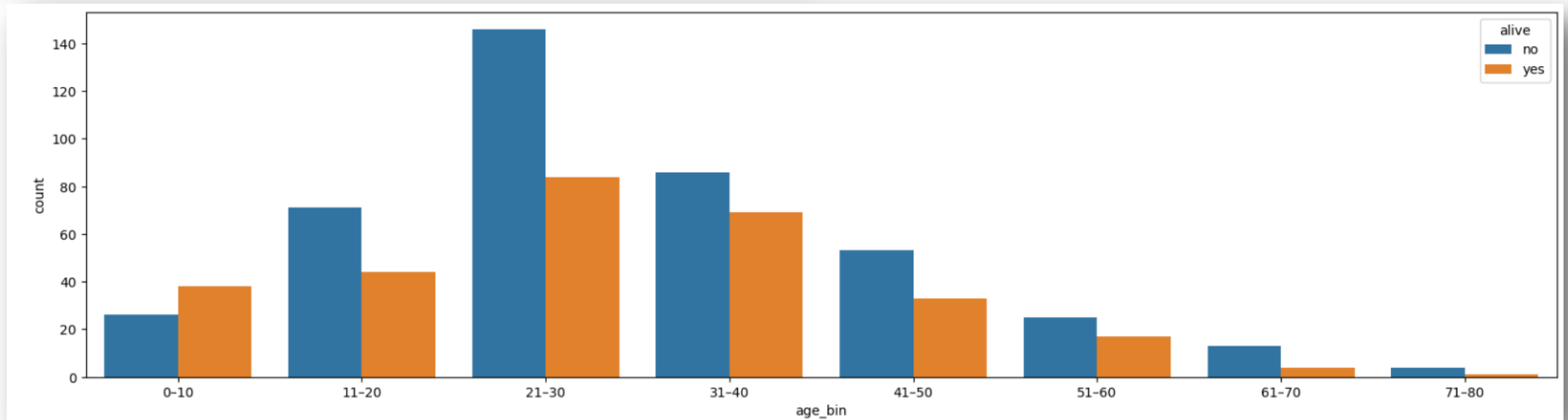
```
bins = [0, 10, 20, 30, 40, 50, 60, 70, 80]
labels = ["0-10", "11-20", "21-30", "31-40", "41-50", "51-60", "61-70", "71-80"]

titanic["age_bin"] = pd.cut(
    titanic["age"],
    bins=bins,
    labels=labels,
    right=True,      # intervals are (a, b]
    include_lowest=True
)

fig = plt.figure(figsize=(20,5))
sns.countplot(x="age_bin", hue="alive", data=titanic)
```



Instead of using the bins like here `titanic['age_bin'] = pd.cut(titanic['age'], 8)`, how can I make my own bins of 0-10, 11-20, etc

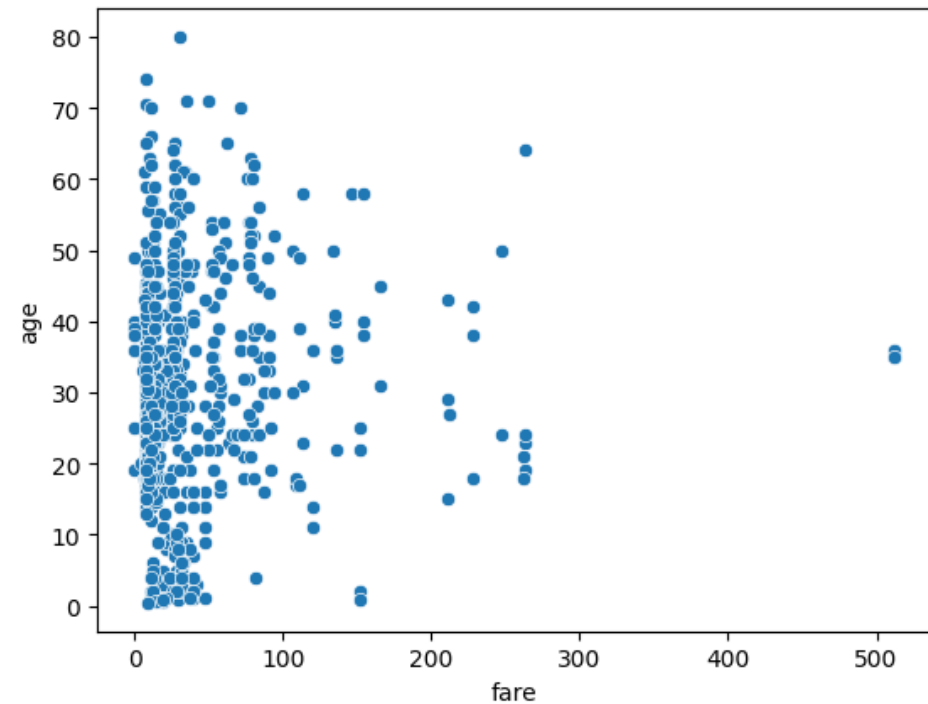


## Bivariate analysis Numerical/Numerical

### Scatter Plots

A **scatter plot** shows the relationship between two continuous variables, making it great for visualizing correlation.

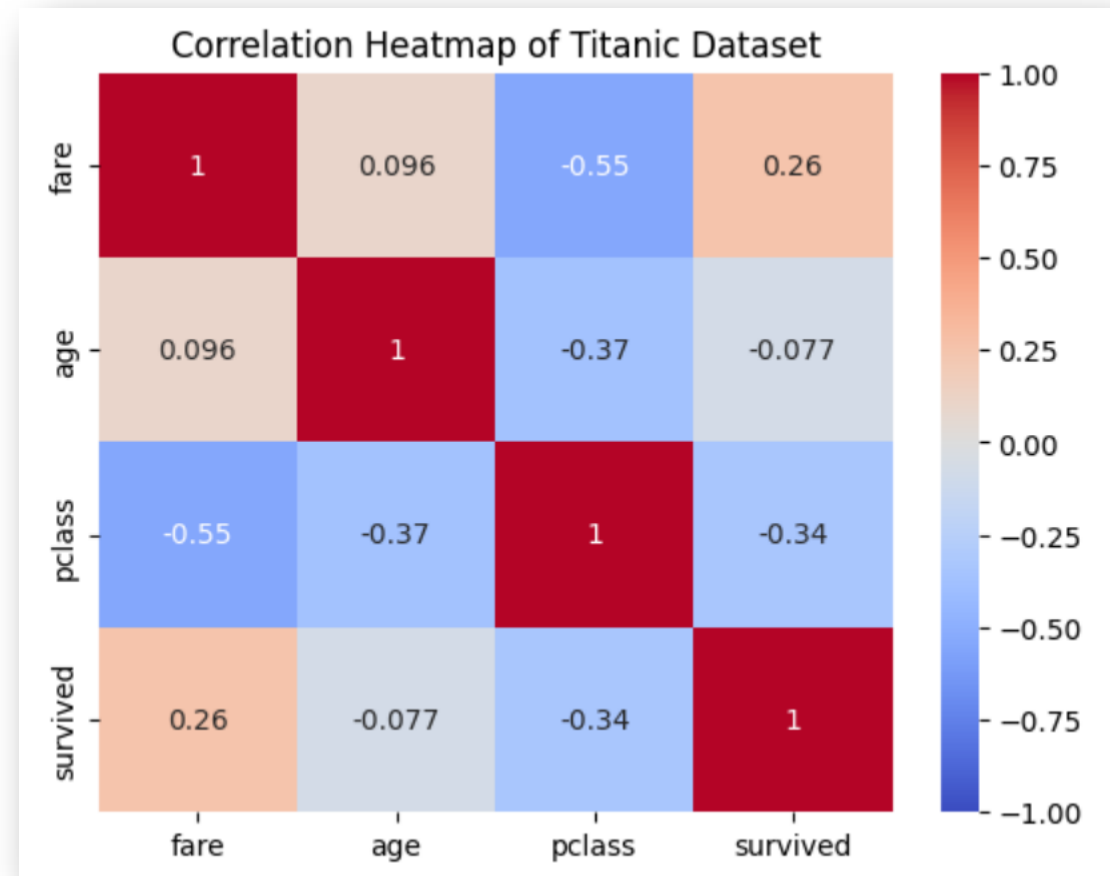
```
sns.scatterplot(x="fare", y="age", data=titanic)
```



```
cols = ["fare", "age", "pclass", "survived"]
corr = titanic[cols].corr()

sns.heatmap(
    corr,
    vmin=-1,
    vmax=1,
    center=0,
    cmap="coolwarm",
    annot=True
)

plt.title("Correlation Heatmap of Titanic Dataset")
plt.show()
```



## Univariate Analysis

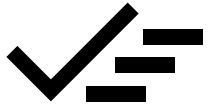


- Numerical feature
- Categorical feature

## Bivariate Analysis



- Numerical/Categorical
- Categorical/Categorical
- Categorical/Numerical
- Categorical/Categorical



## EXPLORATORY DATA ANALYSIS

- Descriptive statistics
- Diagnostic statistics
- Case Study 1 – Titanic

Friday Case Study 2 +  
Examples for Assignment 1